

UNIT – III: Arrays

Introduction, Declaration and Initialization of Arrays, Storage of Array in Computer Memory, Accessing Elements of Arrays.

// java program to perform sum of array ele using 1-d array

```
import java.util.*;
public class SingleDim
{
    public static void main(String args[])
    {
        int size;
        Scanner s=new Scanner(System.in);
        System.out.println("Enter the size of array");
        size=s.nextInt();
        int a[]=new int[size];
        System.out.println("Enter array elements");
        for(int i=0;i<size;i++)
        {
            a[i]=s.nextInt();
        }
        int sum=0;
        for(int i=0;i<size;i++)
        {
            sum=sum+a[i];
        }
        System.out.println("The sum of array ele is"+sum);
    }
}
```

```
/*D:\Openelective2>javac SingleDim.java
```

```
D:\Openelective2>java -cp . SingleDim
```

```
Enter the size of array
```

```
2
```

```
Enter array elements
```

```
10
```

```
20
```

```
The sum of array ele is30*/
```

//java program to perform matrix addition using 2-d array

```
import java.util.*;
public class TwoDim
{
    public static void main(String args[])
    {
        int row1,col1,row2,col2;
        Scanner s=new Scanner(System.in);
        System.out.println("Enter the order of a matrix");
        row1=s.nextInt();
        col1=s.nextInt();
        System.out.println("Enter the order of b matrix");
        row2=s.nextInt();
        col2=s.nextInt();
        if(row1==row2 && col1==col2)
        {
            System.out.println("Matix addition possible");
            System.out.println("Enter the a matrix");
            int a[][]=new int[row1][col1];
            for(int i=0;i<row1;i++)
            {
                for(int j=0;j<col1;j++)
                {
                    a[i][j]=s.nextInt();
                }
            }
            System.out.println("Enter the b matrix");
            int b[][]=new int[row2][col2];
            for(int i=0;i<row2;i++)
            {
                for(int j=0;j<col2;j++)
                {
                    b[i][j]=s.nextInt();
                }
            }
            System.out.println("The resultant matrix is\n");
            for(int i=0;i<row2;i++)
            {
                for(int j=0;j<col2;j++)
                {
                    System.out.print("\t" + (a[i][j]+b[i][j]));
```

```

        }
        System.out.println();
    }
}
else
{
    System.out.println("Matrix addition not possible");
}
}
}
}

```

/*

D:\Openelective2>javac TwoDim.java

D:\Openelective2>java -cp . TwoDim

Enter the order of a matrix

2 3

Enter the order of b matrix

2 3

Matix addition possible

Enter the a matrix

1 0 1

1 1 1

Enter the b matrix

5 2 3

9 6 8

The resultant matrix is

```

    6   2   4
    10  7   9

```

*/

//java program to demonstrate multi dimensional array

```
import java.util.*;
```

```
public class MultiDim
```

```
{
```

```
    public static void main(String args[])
```

```
    {
```

```
        int plane,row,col;
```

```
        System.out.println("Enter plane row and col size");
```

```
        Scanner s=new Scanner(System.in);
```

```
        plane=s.nextInt();
```

```

        row=s.nextInt();
        col=s.nextInt();
        int a[][][]=new int[plane][row][col];
        System.out.println("Enter the multidimensional array elements");
        for(int i=0;i<plane;i++)
        {
            for(int j=0;j<row;j++)
            {
                for(int k=0;k<col;k++)
                {
                    a[i][j][k]=s.nextInt();
                }
            }
        }
        System.out.println("\t"+a[1][1][0]);
    }
}
/*

```

D:\Openelective2>javac MultiDim.java

D:\Openelective2>java -cp . MultiDim

Enter plane row and col size

2

3

3

Enter the multidimensional array elements

2 3 4

5 6 7

8 9 10

11 22 33

44 45 46

47 48 49

44*/

//java program to demonstrate operations of array

```
import java.util.*;
```

```
public class OperationArray
```

```
{
```

```
    public static void main(String args[])
```

```
    {
```

```
        int a[]={10,20,30,40};//static initialization
```

```

        System.out.println("The 2nd array ele is"+a[1]);//accessing
        a[2]=35;//updating
        System.out.println("The array ele are");
        for(int i=0;i<4;i++)//traversing
        {
            System.out.println(a[i]);
        }
        System.out.println("The size of array is"+a.length);
    }
}

```

/*

D:\Openelective2>javac OperationArray.java

D:\Openelective2>java -cp . OperationArray

The 2nd array ele is20

The array ele are

10

20

35

40

The size of array is4*/

//java program to demonstrate single inheritance

```
import java.util.*;
```

```
class Details
```

```
{
```

```
    String name;
```

```
    String roll;
```

```
    String address;
```

```
    Scanner s=new Scanner(System.in);
```

```
    public void getDetails()
```

```
    {
```

```
        System.out.println("Enter the student name ,roll,address");
```

```
        name=s.next();
```

```
        roll=s.next();
```

```
        address=s.next();
```

```
    }
```

```
    public void showDetails()
```

```
    {
```

```
        System.out.println("The Student name , roll,address is");
```

```
        System.out.println("The name is "+name+"\nThe roll is"+roll+"\naddress
is"+address);
    }
}
```

```
class Student extends Details
{
    String branch;
    public void getStudent()
    {
        System.out.println("Enter the student branch");
        branch=s.next();
    }
    public void showStudent()
    {
        System.out.println("The student branch is"+branch);
    }
}
public class SingleInheritance
{
    public static void main(String args[])
    {
        Student s=new Student();
        s.getDetails();
        s.getStudent();
        s.showDetails();
        s.showStudent();
    }
}
```

```
/*D:\Openelective2>javac SingleInheritance.java
```

```
D:\Openelective2>java -cp . SingleInheritance
```

```
Enter the student name ,roll,address
```

```
mahesh
```

```
59
```

```
varanasi
```

```
Enter the student branch
```

```
civil
```

```
The Student name , roll,address is
```

```
The name is mahesh
```

```
The roll is59
```

```
address isvaranasi
```

The student branch iscivil*/

//java program to implement hierarchical inheritance

```
import java.util.*;
class Details
{
    String name,address;
    Scanner s=new Scanner(System.in);
    void getDetails()
    {
        System.out.println("Enter details");
        System.out.println("Enter name");
        name=s.next();
        System.out.println("Enter address");
        address=s.next();
    }
    void showDetails()
    {
        System.out.println("The name and address is");
        System.out.println(name+"\t"+address);
    }
}
class Student extends Details
{
    String roll,branch;
    void getStudent()
    {
        System.out.println("Enter student details");
        System.out.println("Enter roll");
        roll=s.next();
        System.out.println("Enter branch");
        branch=s.next();
    }
    void showStudent()
    {
        System.out.println("The roll and branch is");
        System.out.println(roll+"\t"+branch);
    }
}
class Employee extends Details
```

```

{
    String empid;
    float salary;
    void getEmployee()
    {
        System.out.println("Enter employee details");
        System.out.println("Enter empid");
        empid=s.next();
        System.out.println("Enter salary");
        salary=s.nextFloat();
    }
    void showEmployee()
    {
        System.out.println("The empid and salary is");
        System.out.println(empid+"\t"+salary);
    }
}
public class HierarchicalInheritance
{
    public static void main(String args[])
    {
        Student s=new Student();
        s.getDetails();
        s.getStudent();
        s.showDetails();
        s.showStudent();
        Employee e=new Employee();
        e.getDetails();
        e.getEmployee();
        e.showDetails();
        e.showEmployee();
    }
}

```

/*

D:\Openelective2>javac HierarchicalInheritance.java

D:\Openelective2>java -cp . HierarchicalInheritance

Enter details

Enter name

hari

Enter address

tptyt

Enter student details

Enter roll

12

Enter branch

civil

The name and address is

hari tptyt

The roll and branch is

12 civil

Enter details

Enter name

subbu

Enter address

gdlv

Enter employee details

Enter empid

45

Enter salary

560000

The name and address is

subbu gdlv

The empid and salary is

45 560000.0*/